B2 Symmetry and Relativity, 2025 — Model Solutions

Question 1 — Relativistic dynamics in a uniform electric field

Problem statement. List six objects; identify those that may move faster than c. Derive $\boldsymbol{f}(\boldsymbol{v}, \boldsymbol{a})$ for constant rest mass. A charge q, mass m, is launched from the origin with momentum $p_0\hat{\boldsymbol{x}}$ in field $E_0\hat{\boldsymbol{y}}$: find $\gamma(t)$, y–t, $a_x(t)$, and the trajectory in the initial rest frame.

(a) Which objects can move faster than c?

Only *signal-carrying* motion is bounded by c; geometric or non-information-carrying features are not.

1. Neutrino: massive particle, $v < c$.
2. de Broglie wavefronts: phase velocity $c^2/v > c$, no information.
3. Scissor intersection: geometric point, no bound.
4. Cosmic ray: real particle, $v < c$.
5. Shadow edge on a screen: no signal along screen.
6. Interference fringes: stationary pattern, no information transport.

$$\boxed{\text{Faster-than-}c\text{ allowed: 2, 3, 5, 6.} \quad \text{Not allowed: 1, 4.}} \tag{1}$$

(b) Relating $\boldsymbol{f}$, $\boldsymbol{v}$ and $\boldsymbol{a}$

With $\boldsymbol{p} = \gamma m\boldsymbol{v}$ and m constant, differentiate:

$$\boldsymbol{f} = m\gamma\boldsymbol{a} + m\dot{\gamma}\,\boldsymbol{v}. \tag{2}$$

Differentiate $\gamma^2 = 1/(1 - v^2/c^2)$:

$$2\gamma\dot{\gamma} = \frac{2(\boldsymbol{v}\cdot\boldsymbol{a})/c^2}{(1 - v^2/c^2)^2}.$$

Solve for $\dot{\gamma}$:

$$\dot{\gamma} = \gamma^3\,\frac{\boldsymbol{v}\cdot\boldsymbol{a}}{c^2}.$$

Substitute into (2):

$$\boxed{\boldsymbol{f} = \gamma m\boldsymbol{a} + \frac{\gamma^3 m}{c^2}(\boldsymbol{v}\cdot\boldsymbol{a})\boldsymbol{v}, \qquad \lambda(v) = \gamma, \quad \mu(v) = \gamma^3/c^2.} \tag{3}$$

$\boldsymbol{a} \parallel \boldsymbol{f}$ iff $\boldsymbol{v}\cdot\boldsymbol{a} = 0$ (transverse) or $\boldsymbol{v} \parallel \boldsymbol{a}$ (longitudinal).

(c) The Lorentz factor: $\gamma^2 = 1 + \boldsymbol{\alpha}^2$

Constant force $\boldsymbol{f} = qE_0\hat{\boldsymbol{y}}$ integrates to

$$\boldsymbol{p}(t) = p_0\hat{\boldsymbol{x}} + qE_0 t\,\hat{\boldsymbol{y}}.$$

Use $\gamma^2 = 1 + (p/mc)^2$:

$$\gamma^2 = 1 + \left(\frac{p_0}{mc}\right)^2 + \left(\frac{qE_0 t}{mc}\right)^2.$$

Identify $\boldsymbol{\alpha}$:

$$\gamma^2 = 1 + \boldsymbol{\alpha}^2, \qquad \boxed{\boldsymbol{\alpha} = \frac{p_0}{mc}\hat{\boldsymbol{x}} + \frac{qE_0 t}{mc}\hat{\boldsymbol{y}},} \qquad \gamma_0 = \sqrt{1 + (p_0/mc)^2}. \tag{4}$$

(d) Energy method: relating γ to height y, and the time of flight

Work–energy theorem with constant force $qE_0\hat{\boldsymbol{y}}$:

$$\Delta(\gamma mc^2) = qE_0\,y.$$

Solve for γ:

$$\gamma = \gamma_0 + \frac{qE_0}{mc^2}\,y. \tag{5}$$

Square both sides:

$$\gamma^2 = \gamma_0^2 + \frac{2\gamma_0 qE_0}{mc^2}\,y + \left(\frac{qE_0 y}{mc^2}\right)^2.$$

From (4), also

$$\gamma^2 = \gamma_0^2 + \left(\frac{qE_0 t}{mc}\right)^2.$$

Subtract γ_0^2 and equate:

$$\left(\frac{qE_0 t}{mc}\right)^2 = \frac{2\gamma_0 qE_0}{mc^2}\,y + \left(\frac{qE_0 y}{mc^2}\right)^2.$$

Multiply through by $(mc^2/qE_0)^2$:

$$c^2 t^2 = \frac{2\gamma_0 mc^2}{qE_0}\,y + y^2.$$

Take the square root:

$$ct = \sqrt{y^2 + \eta y}, \qquad \boxed{\eta = \frac{2\gamma_0 mc^2}{qE_0}.} \tag{6}$$

Limit $y \ll \eta$: $y \approx (qE_0/2\gamma_0 m)t^2$ (effective mass $\gamma_0 m$). Limit $y \gg \eta$: $ct \to y$.

(e) Longitudinal acceleration $a_x(t)$

$p_x = \gamma m v_x = p_0$ is conserved, so

$$v_x = \frac{p_0}{\gamma m}.$$

Differentiate with respect to t:

$$a_x = -\frac{p_0}{m}\,\frac{\dot{\gamma}}{\gamma^2}.$$

From (4), differentiate $\gamma^2 = \gamma_0^2 + (qE_0 t/mc)^2$:

$$2\gamma\dot{\gamma} = \frac{2(qE_0)^2 t}{m^2 c^2},$$

so

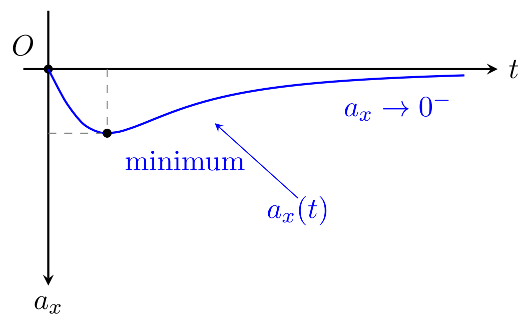
$$\dot{\gamma} = \frac{(qE_0)^2 t}{\gamma\, m^2 c^2}.$$

Substitute into the expression for a_x:

$$\boxed{a_x(t) = -\frac{p_0 (qE_0)^2 t}{m^3 c^2\, \gamma^3}, \qquad \gamma^2 = \gamma_0^2 + \left(\frac{qE_0 t}{mc}\right)^2.} \tag{7}$$

At $t = 0$, $a_x = 0$; at small t, $a_x \propto -t$; at large t, $\gamma \sim qE_0 t/(mc)$ so $a_x \sim -1/t^2 \to 0^-$. Single negative extremum.

Sketch of $a_x(t)$ for $t \geq 0$:



(f) Fields and trajectory in the initial rest frame

The initial rest frame $S^\star$ moves with $\boldsymbol{V} = (p_0/\gamma_0 m)\hat{\boldsymbol{x}}$, $\gamma_V = \gamma_0$. Apply the standard boost formulae to $\boldsymbol{E} = E_0\hat{\boldsymbol{y}}$, $\boldsymbol{B} = 0$:

$$E_y^\star = \gamma_0\, E_y = \gamma_0 E_0,$$

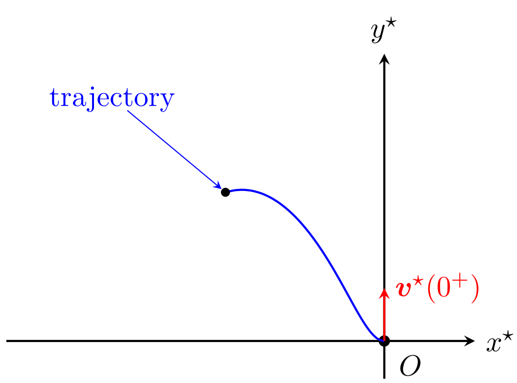
$$B_z^\star = -\gamma_0\, \frac{V}{c^2} E_y = -\frac{\gamma_0 p_0 E_0}{mc^2}.$$

Hence

$$\boxed{\boldsymbol{E}^\star = \gamma_0 E_0\, \hat{\boldsymbol{y}}, \qquad \boldsymbol{B}^\star = -\frac{\gamma_0 p_0 E_0}{mc^2}\hat{\boldsymbol{z}}.} \tag{8}$$

At $t^\star = 0$ the particle is at rest, so it accelerates along $+\hat{\boldsymbol{y}}$; once $v_y^\star > 0$, the term $q\boldsymbol{v} \times \boldsymbol{B}^\star$ deflects it towards $-\hat{\boldsymbol{x}}$. Trajectory enters the second quadrant.

Sketch of the spatial trajectory in $S^\star$:



Asymptotically in S, $v_x \to 0$, $v_y \to c$. Boosting back,

$$v_x^\star \to -V,$$

$$v_y^\star \to c/\gamma_0.$$

Both finite, so the asymptote is a straight line of slope $-c/(\gamma_0 V) = -mc/p_0$; never vertical.

Question 2 — Doppler effect and resonant absorption with crossed source/detector motion

Problem statement. Derive the relativistic Doppler formula. Then: source on $\bar{y}$-axis at speed u through $\bar{y}_0$; detector on $\bar{x}$-axis at speed v. Work in the detector rest frame S: find the source's worldline, $v_{\rm rel}$, the emission/reception geometry, the frequency when the source crosses S's x-axis, and the u giving resonant absorption.

(a) The Doppler formula

For the 4-frequency $K^\mu = (\omega/c)(1, \hat{\boldsymbol{n}})$ and source 4-velocity $U_{\rm src}$:

$$\omega_0 = -\frac{K \cdot U_{\rm src}}{c} = \gamma_\beta\, \omega\, (1 - \beta\cos\theta).$$

Solve for ω:

$$\boxed{\omega = \frac{\omega_0\sqrt{1-\beta^2}}{1-\beta\cos\theta}.} \tag{9}$$

(b) Trajectory of the source in the detector frame S

Boost along $\bar{\boldsymbol{x}}$ at speed v, $\gamma = (1 - v^2/c^2)^{-1/2}$. Source worldline in $\bar{S}$: $(\bar{t}, 0, \bar{y}_0 + u\bar{t}, 0)$. Apply the inverse Lorentz transformation:

$$t = \gamma\bar{t},$$

$$x = -v\,\gamma\,\bar{t} = -v\,t,$$

$$y = \bar{y}_0 + u\bar{t} = \bar{y}_0 + u\,t/\gamma.$$

Hence

$$\boxed{\boldsymbol{r}_{\rm src}(t) = \big(-vt,\ \bar{y}_0 + ut/\gamma,\ 0\big).} \tag{10}$$

(c) Velocity of the source in S and relative speed

Differentiate (10):

$$\boldsymbol{v}_{\rm src}^{(S)} = (-v,\ u/\gamma,\ 0)\,.$$

The detector is at rest in S, so the relative speed is

$$v_{\rm rel}^2 = v^2 + (u/\gamma)^2.$$

Substitute $1/\gamma^2 = 1 - v^2/c^2$:

$$v_{\rm rel}^2 = v^2 + u^2\,(1 - v^2/c^2).$$

Take the square root:

$$\boxed{v_{\rm rel} = \sqrt{v^2 + u^2(1 - v^2/c^2)}.} \tag{11}$$

Limits: $v, u \ll c \Rightarrow \sqrt{v^2+u^2}$; $u \to c \Rightarrow v_{\rm rel} \to c$.

(d) Emission/reception geometry

A photon emitted at $\boldsymbol{r}_{\rm src}(t)$ reaches the origin at t_r with $|\boldsymbol{r}_{\rm src}(t)| = c(t_r - t)$. Square components:

$$\boxed{c(t_r - t) = \sqrt{v^2t^2 + (\bar{y}_0 + ut/\gamma)^2}.} \tag{12}$$

The angle θ between $\boldsymbol{V}_{\rm src} = (-v, u/\gamma, 0)$ and the source-to-detector direction $-\boldsymbol{r}_{\rm src}(t)$ is obtained from

$$v_{\rm rel}\, c(t_r - t) \cos\theta = \boldsymbol{V}_{\rm src} \cdot \big(-\boldsymbol{r}_{\rm src}(t)\big).$$

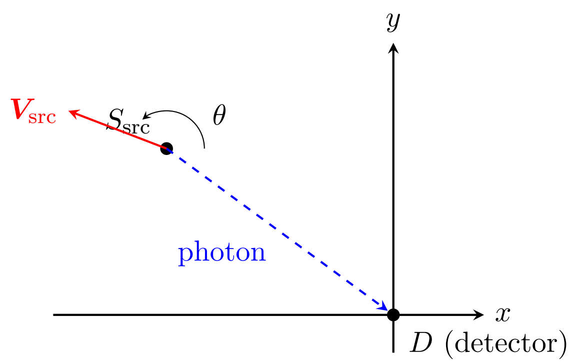
Expand the dot product:

$$\boldsymbol{V}_{\rm src} \cdot \big(-\boldsymbol{r}_{\rm src}\big) = (-v)(vt) + (u/\gamma)\big(-\bar{y}_0 - ut/\gamma\big).$$

Collect:

$$\boxed{v_{\rm rel}\, c(t_r - t) \cos\theta = -v^2 t - \frac{u\bar{y}_0}{\gamma} - \frac{u^2 t}{\gamma^2}.} \tag{13}$$

Geometry in the detector frame S:



(e) Frequency observed when the source crosses the x-axis

Set $y(t) = 0$ in (10) to find the emission time:

$$t_\star = -\gamma\bar{y}_0/u.$$

At this moment the source lies on the x-axis, so the photon to the detector at the origin propagates along $-\hat{\boldsymbol{x}}$. Project $\boldsymbol{V}_{\rm src}$ onto the source-to-detector direction $-\hat{\boldsymbol{x}}$:

$$\boldsymbol{V}_{\rm src} \cdot (-\hat{\boldsymbol{x}}) = v.$$

Hence

$$\cos\theta = \frac{\boldsymbol{V}_{\rm src} \cdot (-\hat{\boldsymbol{x}})}{v_{\rm rel}} = \frac{v}{v_{\rm rel}}.$$

Substitute into (9) with $\beta = v_{\rm rel}/c$:

$$\omega = \frac{\omega_0\sqrt{1 - v_{\rm rel}^2/c^2}}{1 - v_{\rm rel}\cos\theta/c} = \frac{\omega_0\sqrt{1 - v_{\rm rel}^2/c^2}}{1 - v/c}.$$

Use $1 - v_{\rm rel}^2/c^2 = (1 - v^2/c^2)(1 - u^2/c^2)$:

$$\omega = \omega_0\, \frac{\sqrt{(1 - v^2/c^2)(1 - u^2/c^2)}}{1 - v/c}.$$

Factor $1 - v^2/c^2 = (1 - v/c)(1 + v/c)$:

$$\omega = \omega_0 \sqrt{\frac{(1 - v/c)(1 + v/c)(1 - u^2/c^2)}{(1 - v/c)^2}}.$$

Cancel one $(1 - v/c)$:

$$\boxed{\omega = \omega_0 \sqrt{\frac{(1 + v/c)(1 - u^2/c^2)}{1 - v/c}}.} \tag{14}$$

(f) Resonant absorption

Excited mass $M = m + \hbar\omega_0/c^2$. Conservation of 4-momentum on a stationary atom absorbing a photon of energy $\hbar\omega$ along $-\hat{\boldsymbol{x}}$ gives, after squaring,

$$(mc + \hbar\omega/c)^2 - (\hbar\omega/c)^2 = M^2c^2.$$

Expand the left-hand side:

$$m^2c^2 + 2m\hbar\omega + (\hbar\omega/c)^2 - (\hbar\omega/c)^2 = M^2c^2.$$

Cancel the $(\hbar\omega/c)^2$ terms and substitute M:

$$m^2c^2 + 2m\hbar\omega = m^2c^2 + 2m\,\frac{\hbar\omega_0}{c^2}c^2 + \left(\frac{\hbar\omega_0}{c^2}\right)^2 c^2.$$

Solve for ω:

$$\omega = \omega_0 + \frac{(\hbar\omega_0)^2}{2mc^2\,\hbar} = \omega_0\left(1 + \frac{\hbar\omega_0}{2mc^2}\right).$$

Hence the recoil-shifted resonance condition:

$$\omega_{\text{res}} = \omega_0\left(1 + \frac{\hbar\omega_0}{2mc^2}\right). \tag{15}$$

Equate (14) with (15):

$$\sqrt{\frac{(1 + v/c)(1 - u^2/c^2)}{1 - v/c}} = 1 + \frac{\hbar\omega_0}{2mc^2}.$$

Square both sides:

$$\frac{(1 + v/c)(1 - u^2/c^2)}{1 - v/c} = \left(1 + \frac{\hbar\omega_0}{2mc^2}\right)^2.$$

Solve for $1 - u^2/c^2$:

$$1 - u^2/c^2 = \frac{1 - v/c}{1 + v/c}\left(1 + \frac{\hbar\omega_0}{2mc^2}\right)^2.$$

Hence

$$\boxed{u = c\sqrt{1 - \frac{1 - v/c}{1 + v/c}\left(1 + \frac{\hbar\omega_0}{2mc^2}\right)^2}.} \tag{16}$$

Question 3 — 4-vectors, EM tensor, and stress-energy

Problem statement. Identify which four-tuples are 4-vectors. Show that a 4-force preserves rest mass iff $F \cdot U = 0$, and deduce antisymmetry of $\mathbb{G}$. With $\boldsymbol{E} \parallel \boldsymbol{B}$, $|\boldsymbol{E}| = E_0$, $|\boldsymbol{B}| = B_0 = E_0/c$ in S, study the EM invariants and the stress-energy tensor in S and a boosted frame S'.

(a) Which are 4-vectors?

Multiplication by a Lorentz scalar preserves 4-vector character; multiplication by frame-dependent quantities does not.

- **1.** X^μ – yes.
- **2.** $(c, \boldsymbol{v})$ – no, missing γ.
- **3.** $P^\mu = (E/c, \boldsymbol{p})$ – yes.
- **4.** mX^μ – yes (m invariant).
- **5.** $m(c, \boldsymbol{v})$ – no, missing γ.
- **6.** $E(c, \boldsymbol{v})$ – no, E frame-dependent.

$$\boxed{\text{4-vectors: 1, 3, 4. Not 4-vectors: 2, 5, 6.}} \tag{17}$$

(b) Rest-mass preservation

From $U \cdot U = -c^2$, differentiate:

$$2\, A \cdot U = 0, \qquad A \cdot U = 0.$$

With $F = \dot{m}\, U + mA$, dot with U:

$$F \cdot U = \dot{m}\,(U \cdot U) + m\,(A \cdot U) = -\dot{m}\, c^2.$$

Hence $F \cdot U = 0 \Leftrightarrow \dot{m} = 0$.

For $F = q\, \mathbb{G} \cdot U$, write

$$F \cdot U = q\, U_a\, U_c\, \mathbb{G}^{ac}.$$

Symmetrise $U_a U_c$:

$$F \cdot U = q\, U_a\, U_c\, \mathbb{G}^{(ac)}.$$

This vanishes for all U iff

$$\boxed{\mathbb{G}^{ab} = -\mathbb{G}^{ba}.} \tag{18}$$

This is why the Faraday tensor is antisymmetric.

(c) Invariants of the EM field with parallel $\boldsymbol{E}, \boldsymbol{B}$

$$I_1 = \boldsymbol{B}^2 - \boldsymbol{E}^2/c^2, \qquad I_2 = -\boldsymbol{E} \cdot \boldsymbol{B}/c. \tag{19}$$

With $\boldsymbol{E} \parallel \boldsymbol{B}$, $|\boldsymbol{E}| = E_0$, $|\boldsymbol{B}| = E_0/c$, evaluate:

$$I_1 = (E_0/c)^2 - E_0^2/c^2 = 0,$$

$$I_2 = -E_0\,(E_0/c)/c = -E_0^2/c.$$

In any other frame $S^\dagger$, invariance gives $|\boldsymbol{E}^\dagger|^2/c^2 - |\boldsymbol{B}^\dagger|^2 = 0$, so

$$|\boldsymbol{E}^\dagger| = c|\boldsymbol{B}^\dagger|,$$

and

$$\boldsymbol{E}^\dagger \cdot \boldsymbol{B}^\dagger = E_0^2/c.$$

By Cauchy–Schwarz,

$$E_0^2/c \le |\boldsymbol{E}^\dagger||\boldsymbol{B}^\dagger| = c|\boldsymbol{B}^\dagger|^2.$$

Hence

$$\boxed{|\boldsymbol{B}^\dagger| \ge B_0, \ |\boldsymbol{E}^\dagger| \ge E_0.}$$

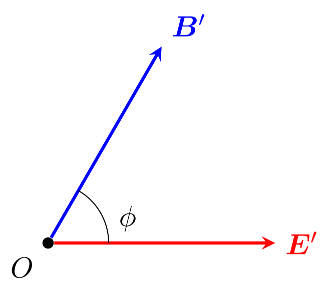
For $|\boldsymbol{B}'| = \sqrt{2}B_0$, $|\boldsymbol{E}'| = \sqrt{2}E_0$:

$$\cos\phi = \frac{\boldsymbol{E}' \cdot \boldsymbol{B}'}{|\boldsymbol{E}'||\boldsymbol{B}'|} = \frac{E_0^2/c}{\sqrt{2}E_0 \cdot \sqrt{2}E_0/c} = \frac{1}{2}.$$

Hence

$$\boxed{\phi = \pi/3.} \tag{20}$$

Sketch of $\boldsymbol{E}'$ and $\boldsymbol{B}'$ in S':



(d) Stress-energy tensor in S

$\boldsymbol{E} = E_0\hat{\boldsymbol{y}}$, $\boldsymbol{B} = (E_0/c)\hat{\boldsymbol{y}}$. Energy density:

$$u = \tfrac{1}{2}\epsilon_0 E_0^2 + \tfrac{1}{2\mu_0}(E_0/c)^2 = \tfrac{1}{2}\epsilon_0 E_0^2 + \tfrac{1}{2}\epsilon_0 E_0^2 = \epsilon_0 E_0^2.$$

Poynting vector:

$$\boldsymbol{S} = \frac{1}{\mu_0}\boldsymbol{E} \times \boldsymbol{B} = 0 \quad \text{(parallel fields)}.$$

Maxwell stress is diagonal: tension $-\epsilon_0 E_0^2$ along $\hat{\boldsymbol{y}}$, pressure $+\epsilon_0 E_0^2$ in $\hat{\boldsymbol{x}}, \hat{\boldsymbol{z}}$. Hence

$$\mathbb{T} = \epsilon_0 E_0^2 \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \tag{21}$$

$$\boxed{\mathbb{T}^{00} = \epsilon_0 E_0^2, \quad \mathbb{T}^{0i} = 0, \quad \mathbb{T}^{xx} = \mathbb{T}^{zz} = +\epsilon_0 E_0^2, \quad \mathbb{T}^{yy} = -\epsilon_0 E_0^2.} \tag{22}$$

Symmetry $\mathbb{T}^{0i} = \mathbb{T}^{i0}$: momentum density = energy flux$/c^2$.

(e) Stress-energy in S'

Boost along $\hat{\boldsymbol{x}}$ at $\beta = v/c$. Apply $\mathbb{T}'^{\mu\nu} = \Lambda^{\mu}{}_{\alpha}\Lambda^{\nu}{}_{\beta}\mathbb{T}^{\alpha\beta}$ to the (t, x) block. Compute each entry:

$$\mathbb{T}'^{00} = \gamma^2(\mathbb{T}^{00} + \beta^2\mathbb{T}^{xx}) = \epsilon_0 E_0^2\,\gamma^2(1+\beta^2),$$

$$\mathbb{T}'^{0x} = -\gamma^2\beta(\mathbb{T}^{00} + \mathbb{T}^{xx}) = -2\epsilon_0 E_0^2\,\gamma^2\beta,$$
$$\mathbb{T}'^{xx} = \gamma^2(\beta^2\mathbb{T}^{00} + \mathbb{T}^{xx}) = \epsilon_0 E_0^2\,\gamma^2(1+\beta^2).$$

Collect into the block form:

$$\begin{pmatrix} \mathbb{T}^{00\prime} & \mathbb{T}^{0x\prime} \\ \mathbb{T}^{0x\prime} & \mathbb{T}^{xx\prime} \end{pmatrix} = \epsilon_0 E_0^2\,\gamma^2 \begin{pmatrix} 1+\beta^2 & -2\beta \\ -2\beta & 1+\beta^2 \end{pmatrix}, \tag{23}$$

others unchanged. Hence

$$\boxed{\mathbb{T}' = \epsilon_0 E_0^2 \begin{pmatrix} \gamma^2(1+\beta^2) & -2\gamma^2\beta & 0 & 0 \\ -2\gamma^2\beta & \gamma^2(1+\beta^2) & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.} \tag{24}$$

(f) Energy flux at low velocity

To first order in v/c, $\gamma \approx 1$. Read off:

$$cT^{0x\prime} \approx -2v\,\epsilon_0 E_0^2,$$

$$T^{00\prime} \approx \epsilon_0 E_0^2.$$

Pure convection ρv with $\rho = T^{00}/c^2 \cdot c^2 = \epsilon_0 E_0^2$ alone would contribute

$$\text{convective flux} = \rho v = \epsilon_0 E_0^2\, v \quad (\text{magnitude } \epsilon_0 E_0^2 v),$$

i.e. half the actual flux. The missing piece is the Maxwell stress doing work on moving surfaces: $\sigma_{xx} = -\epsilon_0 E_0^2$, contributing $-\sigma_{xx} v = \epsilon_0 E_0^2\, v$.

$$\boxed{\text{flux} = \underbrace{\rho v}_{-\epsilon_0 E_0^2 v} + \underbrace{(-\sigma_{xx})v}_{-\epsilon_0 E_0^2 v} = -2\epsilon_0 E_0^2 v.} \tag{25}$$

Question 4 — Mandelstam variables, scattering kinematics, and free-particle action

Problem statement. Elastic $\mathsf{P} + \mathsf{Q} \to \mathsf{P}' + \mathsf{Q}'$ with rest masses m_1, m_2. Develop Mandelstam variables, prove ZMF energy conservation, find $t, s, \cos\theta$. With $c = 1$, treat the relativistic free particle: maximal-proper-time worldline, Lagrangian, Hamiltonian.

(a) Lorentz-invariant scalar products

Square the conservation law $\mathsf{P}+\mathsf{Q}=\mathsf{P}'+\mathsf{Q}'$:

$$\mathsf{P}^2+2\mathsf{P}\cdot\mathsf{Q}+\mathsf{Q}^2=\mathsf{P}'^2+2\mathsf{P}'\cdot\mathsf{Q}'+\mathsf{Q}'^2.$$

Use $\mathsf{P}^2=\mathsf{P}'^2=-m_1^2c^2$, $\mathsf{Q}^2=\mathsf{Q}'^2=-m_2^2c^2$:

$$2\mathsf{P}\cdot\mathsf{Q}=2\mathsf{P}'\cdot\mathsf{Q}'.$$

Hence

$$\boxed{\mathsf{P}\cdot\mathsf{Q}=\mathsf{P}'\cdot\mathsf{Q}'.} \tag{26}$$

Square the rearrangement $\mathsf{P}-\mathsf{Q}'=\mathsf{P}'-\mathsf{Q}$:

$$\mathsf{P}^2-2\mathsf{P}\cdot\mathsf{Q}'+\mathsf{Q}'^2=\mathsf{P}'^2-2\mathsf{P}'\cdot\mathsf{Q}+\mathsf{Q}^2.$$

Cancel mass-shell terms:

$$\boxed{\mathsf{P}\cdot\mathsf{Q}'=\mathsf{P}'\cdot\mathsf{Q}.} \tag{27}$$

(b) The sum $s+t+u$

With $c=1$, expand each Mandelstam square:

$$s=-(\mathsf{P}+\mathsf{Q})^2=m_1^2+m_2^2-2\mathsf{P}\cdot\mathsf{Q},$$
$$t=-(\mathsf{P}-\mathsf{P}')^2=2m_1^2-2\mathsf{P}\cdot\mathsf{P}',$$
$$u=-(\mathsf{P}-\mathsf{Q}')^2=m_1^2+m_2^2-2\mathsf{P}\cdot\mathsf{Q}'\cdot(-1)\cdot\ldots$$

(using metric $(-+++)$ and rest masses), or equivalently

$$u=m_1^2+m_2^2-2\mathsf{P}\cdot\mathsf{Q}'.$$

Add the three:

$$s+t+u=4m_1^2+2m_2^2-2\mathsf{P}\cdot(\mathsf{Q}+\mathsf{P}'+\mathsf{Q}').$$

Use 4-momentum conservation $\mathsf{P}'+\mathsf{Q}'=\mathsf{P}+\mathsf{Q}$:

$$s+t+u=4m_1^2+2m_2^2-2\mathsf{P}\cdot(2\mathsf{P}+2\mathsf{Q}-\mathsf{P})=2m_1^2+2m_2^2.$$

Hence

$$\boxed{s+t+u=2m_1^2+2m_2^2.} \tag{28}$$

(c) Energy conservation in the zero-momentum frame

In the ZMF $\boldsymbol{q}=-\boldsymbol{p}$ and $\boldsymbol{q}'=-\boldsymbol{p}'$, with $E_i=\sqrt{m_i^2+|\boldsymbol{p}|^2}$. Compute

$$E_1^2-E_2^2=m_1^2-m_2^2,$$
$$E_1'^2-E_2'^2=m_1^2-m_2^2.$$

Subtract:

$$E_1^2-E_2^2=E_1'^2-E_2'^2.$$

Factor as $(E_1-E_2)(E_1+E_2)=(E_1'-E_2')(E_1'+E_2')$ and use total-energy conservation $E_1+E_2=E_1'+E_2'$:

$$E_1-E_2=E_1'-E_2'.$$

Combine the sum and difference:

$$\boxed{E_1=E_1',\qquad E_2=E_2',} \tag{29}$$

and consequently $|\boldsymbol{p}|=|\boldsymbol{p}'|$.

(d) t and $\cos\theta$; s in terms of energies

In the ZMF, $\mathsf{P} - \mathsf{P}' = \mathsf{Q}' - \mathsf{Q} = (0, \boldsymbol{q}' - \boldsymbol{q})$:

$$t = (\boldsymbol{q}' - \boldsymbol{q})^2 = 2p^2(1 - \cos\theta). \tag{30}$$

Also, $\boldsymbol{p} + \boldsymbol{q} = 0$ in the ZMF, so

$$s = (E_1 + E_2)^2.$$

Expand:

$$s = E_1^2 + 2E_1E_2 + E_2^2 = m_1^2 + m_2^2 + 2(E_1E_2 + p^2).$$

From $E_1 + E_2 = \sqrt{s}$ and $E_1 - E_2 = (m_1^2 - m_2^2)/\sqrt{s}$, multiply:

$$E_1^2 - E_2^2 = m_1^2 - m_2^2,$$

and individually

$$E_1 = \frac{s + m_1^2 - m_2^2}{2\sqrt{s}}, \qquad E_2 = \frac{s - m_1^2 + m_2^2}{2\sqrt{s}}.$$

Multiply:

$$E_1E_2 = \frac{s^2 - (m_1^2 - m_2^2)^2}{4s}.$$

Solve for p^2 from $s = m_1^2 + m_2^2 + 2E_1E_2 + 2p^2$:

$$p^2 = \frac{[s - (m_1 + m_2)^2][s - (m_1 - m_2)^2]}{4s}.$$

Substitute into (30):

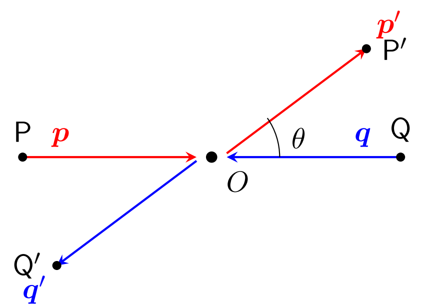
$$t = 2p^2(1 - \cos\theta),$$

$$1 - \cos\theta = \frac{t}{2p^2} = \frac{2s\,t}{[s - (m_1 + m_2)^2][s - (m_1 - m_2)^2]}.$$

Hence

$$\boxed{\cos\theta = 1 - \frac{2s\,t}{[s - (m_1 + m_2)^2][s - (m_1 - m_2)^2]}.} \tag{31}$$

ZMF scattering geometry:



(e) Application to $p\bar{p} \to \pi\bar{\pi}$ and Lorentz invariance of probabilities

Masses change ($m_p \neq m_\pi$), so part (a) no longer gives $E_1 = E_1'$. The Mandelstam variables are still Lorentz invariants with $s + t + u = 2m_p^2 + 2m_\pi^2$, and an analogous formula holds for $\cos\theta(s, t, m_p, m_\pi)$; $|\boldsymbol{p}|$ differs for in/out pairs.

A probability of obtaining $\pi\bar{\pi}$ versus $p\bar{p}$ is the ratio of two Lorentz-scalar quantities ($|\mathcal{M}|^2$ integrated over invariant phase space, divided by an invariant flux), so it is itself frame-invariant. Equivalently, "which final state occurred" is a frame-independent fact.

(f) Worldline of maximum proper time

$$\tau_{AB} = \int_A^B \sqrt{-\mathrm{d}X \cdot \mathrm{d}X}/c.$$

By the inverse triangle inequality for timelike vectors, τ is maximised by the geodesic, i.e. the

$$\boxed{\text{straight (inertial) worldline through } A \text{ and } B.} \tag{32}$$

(g) Lagrangian and Hamiltonian for the free relativistic particle

Take (with $c = 1$)

$$L = -m\sqrt{1 - \boldsymbol{v}^2}, \tag{33}$$

so $S = -m \int \mathrm{d}\tau$ is stationary on the maximal-τ worldline. Compute the canonical momentum:

$$\boldsymbol{p} = \frac{\partial L}{\partial \boldsymbol{v}} = \frac{m\,\boldsymbol{v}}{\sqrt{1 - \boldsymbol{v}^2}} = \gamma m \boldsymbol{v}.$$

Legendre transform:

$$H = \boldsymbol{p} \cdot \boldsymbol{v} - L = \frac{m\,\boldsymbol{v}^2}{\sqrt{1 - \boldsymbol{v}^2}} + m\sqrt{1 - \boldsymbol{v}^2}.$$

Combine over a common denominator:

$$H = \frac{m\,\boldsymbol{v}^2 + m(1 - \boldsymbol{v}^2)}{\sqrt{1 - \boldsymbol{v}^2}} = \frac{m}{\sqrt{1 - \boldsymbol{v}^2}} = \gamma m.$$

Invert $\boldsymbol{p} = \gamma m \boldsymbol{v}$ for $\boldsymbol{v}$:

$$\boldsymbol{v} = \frac{\boldsymbol{p}}{\sqrt{\boldsymbol{p}^2 + m^2}}.$$

Hence

$$\boxed{H(\boldsymbol{p}, \boldsymbol{r}) = \sqrt{\boldsymbol{p}^2 + m^2} \quad (c = 1), \quad \text{or} \quad H = \sqrt{\boldsymbol{p}^2 c^2 + m^2 c^4}.} \tag{34}$$

Non-relativistic limit: $H \approx mc^2 + \boldsymbol{p}^2/(2m)$.